CFRM 502, Winter 2020.

Homework 2

Due: Wednesday 22 January 2020, NO LATER than 11:59pm

Instructions: You must submit your work through Canvas, and in the form of two attachments: A .pdf file of your detailed answers and a file containing your codes (.R or .Rmd). The .pdf files can be typeset or be a readable scan of your handwritten notes. You do not need to include R outputs in the .pdf file. However, you must clearly indicate the relevant part of your R codes when answering a question that is based on your code.

The maximum number of points you can receive for this homework is 35.

1. Consider the mutual fund "AMRMX" and 4 exchange traded funds (ETFs) with the symbols "IWD", "IWF", "IWO", and "VONG". Use the R package "quantmod" (using Yahoo! Finance as the source) to obtain price series from January 3, 2011 to December 29, 2016.

   (a) (2 pts) Compute the daily log returns of the 5 symbols. Produce a scatter plot and a correlation plot. Interpret the plots.

   (b) (2 pts) Fit a regression model with AMRMX log returns as the response and the ETFs' log returns as predictors. State the value of the coefficients of the regression. Are the coefficients significant under 5% significance level? What is the sample correlation between fitted and observed values of AMRMX log returns?

   (c) (2 pts) Obtain the ANOVA table for the model in part (b), and interpret it. What is the total sum of squares of the model?

   (d) (2 pts) Do a F-test to see if the coefficients of "IWO" and "VONG" and the intercept are simultaneously zero. That is, the null hypothesis is $H_0 : \beta_{\text{IWO}} = \beta_{\text{VONG}} = \beta_0 = 0$.

   (e) (2 pts) If you can only choose one ETF (and the intercept) as the predictor, which one would you choose? What if you can choose 2 predictors? What is the best model with three predictors? You must support your answers using appropriate R outputs.

   (f) (2 pts) Find the best set of predictors based on BIC criterion.

   (g) (2 pts) Do you see any evidence of collinearity between the predictors? Justify your answer by an appropriate R output.

2. (3 pts) Solve exercise 9, in section 9.12 of the textbook. Explain your answers.

3. The forward-rate curve is an important tool for modeling the term-structure of interest rates. It is given by

$$r(t) = -\frac{\partial}{\partial t} \log P(t),$$

   where $P(t)$ is the price of a zero-coupon bond with maturity t. The forward-rate curve cannot be directly observed from the market, and one way to estimate it is by using the prices of zero-coupon bonds with different maturities. Specifically, let $P(t_0), P(t_1), \ldots, P(t_n)$ be the prices of $n+1$ zero-coupon bonds. Then, $r(t_i)$, the forward-rate curve at t_i, is estimated by the "empirical forward-rate estimates" $\widehat{r}_i$ given by

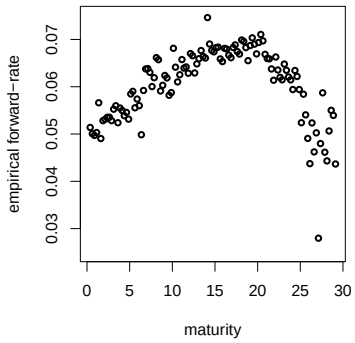
$$\widehat{r}_i := -\frac{\log P(t_i) - \log P(t_{i-1})}{t_i - t_{i-1}}; \quad i \in \{1, \ldots, n\}.$$

These estimates are noisy. To obtain a forward rate, one needs to fit a "curve" through the points $(t_1, \widehat{r}_1), \ldots, (t_n, \widehat{r}_n)$. This will be our main goal for this problem.

The text file "HW2-data.txt" that is posted with this homework, includes the prices of 117 zero-coupon bonds with equally spaced maturities from 0 to 30 years. The following code snippet loads the data and stores the empirical forward-rate estimates and the corresponding maturities in the variables "EMP" and "MAT", respectively. When running the code, make sure that the text file is in your working directory.

```
dat = read.table("HW2-data.txt",header=TRUE)
dat = dat[order(dat$T),]
EMP = -diff(log(dat$price))/diff(dat$T)
MAT = dat$T[-1]
```

The following graph shows a scatterplot of the empirical forward-rate (EMP) vs. maturity (MAT). Clearly, the forward rate curve is not linear.



(a) (1 pts) Ignoring the nonlinearity, we decide to fit a line regression $\text{EMP}_i = \beta_0 + \beta_1 \text{MAT}_i + \varepsilon_i$ to the data. Fit this model (i.e. express the estimated parameters) and provide a scatter plot of the predictor and the response along with the fitted line.

(b) (3 pts) Check the fitted model in part (a) for the following anomalies: high-leverage points, residual outliers, and high-influence points. Explain your answers and provide necessary plots to justify them.

(c) (4 pts) Check the model in part (a) for non-normality, non-constant variance of residuals, non-linear relation between response and predictor, as well as spurious regression. Explain your answers and provide supporting plots.

(d) (2 pts) Not satisfied with the model in part (a), we decide to use a cubic regression model

$$\text{EMP}_i = \beta_0 + \beta_1 \,\text{MAT}_i + \beta_2 \,\text{MAT}_i^2 + \beta_3 \,\text{MAT}_i^3 + \varepsilon_i.$$

Fit this model using a multiple linear regression model. Specify the estimated parameters $\widehat{\beta}_0, \ldots, \widehat{\beta}_3$ and provide a scatter plot of the EMP versus MAT along with the fitted curve. You are only allowed to use the R function "lm()" to fit the model.

(e) (7 pts) Repeat parts (b) and (c) for the model in part (d).

(f) (1 pts) Can you detect collinearity (between predictors) in the model in part (d)?